

The Environmental Impact of the Silicon needed to produce AI models. Thoughts?

Posted by u/NamelessGeo • • 0 points • 1 comments

Saw this was conceptualized as a place for debate so ill throw my two cents in. My primary issue with ai is an environmental one.

Lots of folks talk about the energy cost and that is substantial in training but it's also a non-issue as you can just power it using renewable or nuclear sources. And actually generating something is minuscule, only slightly more energy than a google search

The demand for massive graphics card rigs however is not so easily sidestepped. To run the models that power ai tools like chat gpt and midjourney you need gpus and batteries. lots of them. So big bulk gpu orders from major companies trying to build their own models have come in droves (as can be seen from nvidias ludicrously out of proportion stock price).

A core component of electronics manufacturing is silicon which has had its production massively increased since the advent of AI technology at its current level. From 4.31 million metric tons in 2022 to 9.7 million metric tons in 2024.

The reason this is a problem is because silicon has a 3 pronged environmental impact:

* ***"Production Process:** The creation of silicone relies on hydrocarbons derived from petroleum, a non-renewable resource. This process is energy-intensive and contributes to carbon emissions.

* **Recyclability:** Silicone is notoriously difficult to recycle. Most curbside recycling programs do not accept it, and only a few speciality recyclers handle silicone, limiting its end-of-life options.

* ***Biodegradability:** Silicone is not biodegradable or compostable within a human lifetime. While it may break down into smaller pieces over time, it does not decompose fully, posing long-term environmental risks."

[<https://www.greenmatch.co.uk/blog/is-silicone-bad-for-the-environment>]

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How do you think, people can avoid or atleast mitigate the amount this aspect of the production process needed to keep these things running at the current level? Should there be permits implemented on who can assemble rigs? Should we be funding technology to

develop gpus with more sustainable components? Such as quantum computing, organic semiconductors (although i know that tech has been dead in the water since the Jan Hendrik Schön scandal) etc... Would love to hear you all's thoughts.

Comments

u/YentaMagenta • 2 points • 2025-05-25 06:05 UTC

Oh honey, sweetie, baby... Silicone is a rubbery material used for caulking, ice cube trays, high quality dildos and so much more, but it's not what you use to make computer chips. That's **silicon** with no 'e' on the end.

And silicon may well have environmental impacts but everything in your post seems focused on silicone.

So maybe you should delete this, do some more research, and get back to us when you understand the difference between the two materials.

u/NamelessGeo • 1 points • 2025-05-25 06:11 UTC

BRB killing myself.